Embedded System Lab Manual

Subject Name: Embedded System Lab

Subject Code: 7IT06

Experiment No.: 3

Title: Write and upload a basic LED blinking program using the C language on an Arduino Uno board.

Objectives:

1. To understand the concept of digital output control using Arduino Uno.
2. To write and upload a basic C program to blink an LED at a fixed time interval.

Required Components/Kit:

- Arduino Uno Board
- USB Cable
- Computer with Arduino IDE Installed
- 1 x LED
- 1 x 220Ω Resistor
- Breadboard and Jumper Wires

Theory:

The **Arduino Uno** is a microcontroller board based on the **ATmega328P**. It has 14 digital input/output pins, and digital pin **13** is commonly connected to the onboard LED.

The LED blinking program is a basic program that demonstrates how to control a digital output using software. It helps students understand:

- GPIO (General Purpose Input Output)
- Use of pinMode(), digitalWrite(), and delay() functions

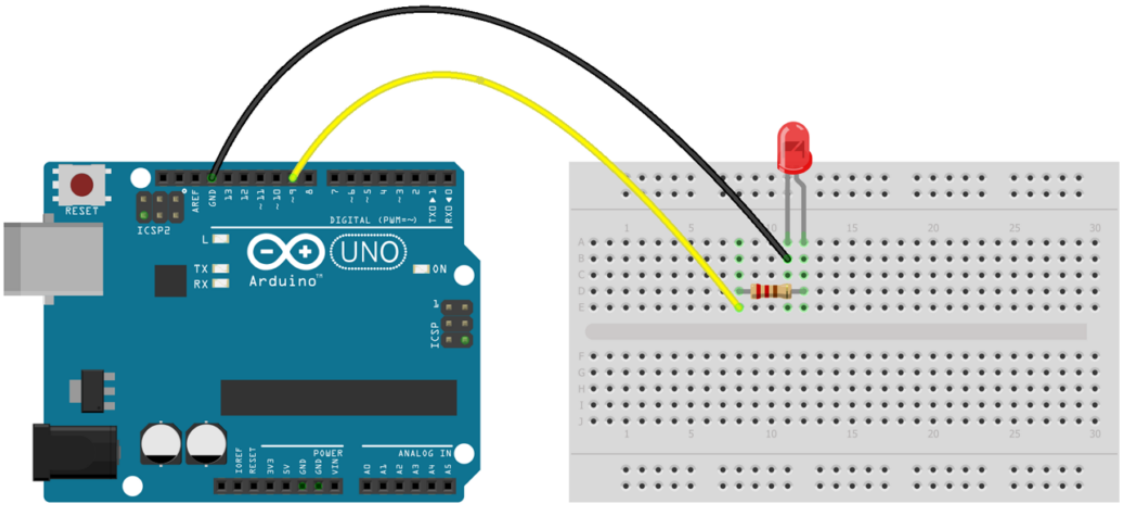
Procedure:

1. **Connect the Arduino Uno** to the computer using a USB cable.
2. **Open Arduino IDE** on your computer.

3. **Write the following C code** in the IDE:
4. Click **"Verify"** (✓ icon) to compile the code.
5. Select the **correct COM port** and **board (Arduino Uno)** from Tools menu.
6. Click **"Upload"** (→ icon) to upload the code to the board.
7. Observe the **onboard LED (pin 13)** blinking with a 1-second interval.

Circuit Diagram (Description)

1. Connect LED anode to pin 13 of Arduino Uno via 220Ω resistor.
2. Connect LED cathode to GND.



Code:

```
void setup()
{
        pinMode(13, OUTPUT);                    // Set pin 13 as output
}
void loop()
{
        digitalWrite(13, HIGH);                 // Turn the LED on
        delay(1000);                            // Wait for 1 second (1000 ms)
        digitalWrite(13, LOW);                  // Turn the LED off
        delay(1000);                            // Wait for 1 second
}
```

Working Principle:

- When digital pin is set HIGH → LED turns ON.
- When digital pin is set LOW → LED turns OFF.
- Repeated alternation with delays makes the LED blink.

Observation Table:

Sr No	Action Performed	Expected Result
1	Upload code to Arduino	LED on pin 13 starts blinking
2	Change delay to 500 ms	LED blinks faster (every 0.5 sec)

Result:

The basic LED blinking program was successfully written and uploaded using the Arduino Uno and C language. The onboard LED blinked at 1-second intervals, verifying digital output control.

Conclusion:

This experiment demonstrated the basic structure of an embedded C program and how to control an output device (LED) using an Arduino microcontroller.